

Differential Geometry

Solved Past Paper

Unit Speed Curves • Frenet–Serret Apparatus • Non-Unit Speed Curves

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Class: BS Part IV
Session: 1st Semester Examination 2025
Date: Thursday, 22nd May 2025
Total Marks: 50 (Attempt any **Four**, two from each section)
Time: 2 Hours

Section A – Unit Speed Curves

Q.No. 1

Regular Curves and Re-parametrization

(a) Define a regular curve and a re-parametrization of a curve.

Solution

A curve $\alpha : (a, b) \rightarrow \mathbb{R}^3$ is **regular** if $\alpha'(t) \neq \mathbf{0}$ for all $t \in (a, b)$.

A **re-parametrization** is a diffeomorphism $g : (c, d) \rightarrow (a, b)$ (smooth bijection with $g' \neq 0$). The curve $\beta = \alpha \circ g$ is then a re-parametrization of α .

(b) Let $\beta = \alpha \circ g$, $t_0 = g(s_0)$. Show that the unit tangent S of β at s_0 satisfies $S = \pm T$, where T is the unit tangent of α at t_0 .

Solution

Step 1. Apply the chain rule to $\beta = \alpha \circ g$:

$$\beta'(s_0) = \alpha'(g(s_0)) \cdot g'(s_0) = \alpha'(t_0) \cdot g'(s_0).$$

Step 2. Find the unit tangent of β :

$$S = \frac{\beta'(s_0)}{|\beta'(s_0)|} = \frac{\alpha'(t_0) \cdot g'(s_0)}{|\alpha'(t_0)| \cdot |g'(s_0)|} = \frac{g'(s_0)}{|g'(s_0)|} \cdot T = \pm T.$$

Conclusion: $S = +T$ if $g'(s_0) > 0$ (same direction), $S = -T$ if $g'(s_0) < 0$ (opposite direction). $\square$

(c) Show that α and $\beta = \alpha \circ g$ have the same arc length.

Solution

Step 1. Write the arc length of α :

$$L(\alpha) = \int_a^b |\alpha'(t)| dt.$$

Step 2. Substitute $t = g(s)$, so $dt = g'(s) ds$. When t goes from a to b , s goes from c to d :

$$L(\alpha) = \int_c^d |\alpha'(g(s))| |g'(s)| ds = \int_c^d |\alpha'(g(s)) \cdot g'(s)| ds = \int_c^d |\beta'(s)| ds = L(\beta).$$

Conclusion: $L(\alpha) = L(\beta)$. Arc length does not depend on parametrization. $\square$

Q.No. 2

Torsion and Curvature of a Plane Curve

(a) Let $\alpha(s)$ be a C^4 unit-speed curve in the (x, y) -plane with $\kappa \neq 0$. Prove $\tau \equiv 0$.

Solution

Step 1. Write $\alpha(s) = (x(s), y(s), 0)$. Compute the Frenet vectors:

$$T = \alpha' = (x', y', 0), \quad \kappa N = \alpha'' = (x'', y'', 0).$$

Step 2. Compute the binormal $B = T \times N$:

$$B = \frac{1}{\kappa} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x' & y' & 0 \\ x'' & y'' & 0 \end{vmatrix} = \frac{1}{\kappa} (0, 0, x'y'' - y'x'') \cdot \hat{k} = \pm(0, 0, 1) = \pm \mathbf{k}.$$

Step 3. Since the curve lies in the $z = 0$ plane, $B = \pm \mathbf{k}$ is a *constant* vector. So:

$$B'(s) = \mathbf{0} \implies -\tau N = \mathbf{0} \implies \tau = 0 \quad (\text{since } \kappa \neq 0 \Rightarrow N \neq \mathbf{0}). \quad \square$$

(b) Let $\alpha(s) = (x(s), y(s), 0)$ be unit-speed. Prove $\kappa = |x'y'' - x''y'|$.

Solution

Step 1. Since α is unit-speed: $|\alpha'| = 1$, so

$$x'^2 + y'^2 = 1.$$

Step 2. Differentiate the above:

$$x'x'' + y'y'' = 0.$$

Step 3. Apply the Lagrange identity $|a \times b|^2 = |a|^2|b|^2 - (a \cdot b)^2$ with $a = (x', y')$, $b = (x'', y'')$:

$$|x'y'' - x''y'|^2 + (x'x'' + y'y'')^2 = (x'^2 + y'^2)(x''^2 + y''^2).$$

Step 4. Substitute Steps 1 and 2:

$$|x'y'' - x''y'|^2 + 0 = 1 \cdot (x''^2 + y''^2) = |\alpha''|^2 = \kappa^2.$$

Therefore $\kappa = |x'y'' - x''y'|$. $\square$

Q.No. 3

Osculating Planes at $s = 0$ [Equation: $(\mathbf{r} - \alpha(0)) \cdot B(0) = 0$]

$$(a) \alpha(s) = \left(\frac{5}{13} \cos s, \frac{8}{13} - \sin s, -\frac{12}{13} \cos s \right)$$

Solution

$$\alpha'(s) = \left(-\frac{5}{13} \sin s, -\cos s, \frac{12}{13} \sin s \right), \quad |\alpha'|^2 = \frac{25}{169} \sin^2 s + \cos^2 s + \frac{144}{169} \sin^2 s = 1. \checkmark$$

At $s = 0$:

$$\alpha(0) = \left(\frac{5}{13}, \frac{8}{13}, -\frac{12}{13} \right), \quad T = (0, -1, 0), \quad \alpha''(0) = \left(-\frac{5}{13}, 0, \frac{12}{13} \right) \Rightarrow \kappa = 1, \quad N = \left(-\frac{5}{13}, 0, \frac{12}{13} \right).$$

$$B = T \times N = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -1 & 0 \\ -\frac{5}{13} & 0 & \frac{12}{13} \end{vmatrix} = \left(-\frac{12}{13}, 0, -\frac{5}{13} \right).$$

Osculating plane:

$$-\frac{12}{13}(x - \frac{5}{13}) - \frac{5}{13}(z + \frac{12}{13}) = 0 \implies \boxed{12x + 5z = 0}.$$

$$(b) \alpha(s) = \frac{1}{\sqrt{5}}(\sqrt{1+s^2}, 2s, \ln(s + \sqrt{1+s^2}))$$

Solution

$$\alpha'(s) = \frac{1}{\sqrt{5}} \left(\frac{s}{\sqrt{1+s^2}}, 2, \frac{1}{\sqrt{1+s^2}} \right), \quad |\alpha'|^2 = \frac{1}{5} \left(\frac{s^2+1}{1+s^2} + 4 \right) = 1. \checkmark$$

At $s = 0$:

$$\alpha(0) = \left(\frac{1}{\sqrt{5}}, 0, 0 \right), \quad T = \left(0, \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right), \quad \alpha''(0) = \left(\frac{1}{\sqrt{5}}, 0, 0 \right) \Rightarrow \kappa = \frac{1}{\sqrt{5}}, \quad N = (1, 0, 0).$$

$$B = T \times N = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ 1 & 0 & 0 \end{vmatrix} = \left(0, \frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}} \right).$$

Osculating plane:

$$\frac{y}{\sqrt{5}} - \frac{2z}{\sqrt{5}} = 0 \implies \boxed{y = 2z}.$$

Section B – Non-Unit Speed Curves

Q.No. 4

Frenet Formulas for a Regular (Non-Unit Speed) Curve $\beta(t)$ in $\mathbb{R}^3$

Let $\dot{\beta}, \ddot{\beta}, \dddot{\beta}$ denote derivatives w.r.t. t , and $\dot{s} = |\dot{\beta}|$. Key expansion: $\dot{\beta} = \dot{s}T$, $\ddot{\beta} = \ddot{s}T + \dot{s}^2\kappa N$, giving $\dot{\beta} \times \ddot{\beta} = \dot{s}^3\kappa B$.

(i) Show $B = \frac{\dot{\beta} \times \ddot{\beta}}{|\dot{\beta} \times \ddot{\beta}|}$.

Solution

Step 1. Express $\dot{\beta}$ and $\ddot{\beta}$ in the Frenet frame. Since $\dot{s} = |\dot{\beta}|$:

$$\dot{\beta} = \dot{s}T, \quad \ddot{\beta} = \ddot{s}T + \dot{s}^2\kappa N.$$

Step 2. Take the cross product (using $T \times T = 0$ and $T \times N = B$):

$$\dot{\beta} \times \ddot{\beta} = (\dot{s}T) \times (\ddot{s}T + \dot{s}^2\kappa N) = \dot{s}\ddot{s}(T \times T) + \dot{s}^3\kappa(T \times N) = \dot{s}^3\kappa B.$$

Step 3. Divide by its magnitude ($|B| = 1$, so $|\dot{\beta} \times \ddot{\beta}| = \dot{s}^3\kappa$):

$$\frac{\dot{\beta} \times \ddot{\beta}}{|\dot{\beta} \times \ddot{\beta}|} = \frac{\dot{s}^3\kappa B}{\dot{s}^3\kappa} = B. \quad \square$$

(ii) Show $\kappa = \frac{|\dot{\beta} \times \ddot{\beta}|}{|\dot{\beta}|^3}$.

Solution

Step 1. From part (i): $|\dot{\beta} \times \ddot{\beta}| = \dot{s}^3\kappa$.

Step 2. Since $\dot{s} = |\dot{\beta}|$, we have $\dot{s}^3 = |\dot{\beta}|^3$. So:

$$|\dot{\beta} \times \ddot{\beta}| = |\dot{\beta}|^3\kappa \implies \kappa = \frac{|\dot{\beta} \times \ddot{\beta}|}{|\dot{\beta}|^3}. \quad \square$$

(iii) Show $\tau = \frac{[\dot{\beta}, \ddot{\beta}, \dddot{\beta}]}{|\dot{\beta} \times \ddot{\beta}|^2}$.

Solution

Step 1. Differentiate $\ddot{\beta} = \ddot{s}T + \dot{s}^2\kappa N$ using $\dot{N} = \dot{s}(-\kappa T + \tau B)$:

$$\ddot{\beta} = (\ddot{s} - \dot{s}^3\kappa^2)T + (\text{some coefficient})N + \dot{s}^3\kappa\tau B.$$

The key point: the B -component of $\ddot{\beta}$ is $\dot{s}^3\kappa\tau$.

Step 2. Write all three in the $\{T, N, B\}$ frame and take the triple product:

$$[\dot{\beta}, \ddot{\beta}, \ddot{\beta}] = \det \begin{pmatrix} \dot{s} & 0 & 0 \\ \ddot{s} & \dot{s}^2\kappa & 0 \\ * & * & \dot{s}^3\kappa\tau \end{pmatrix} = \dot{s} \cdot \dot{s}^2\kappa \cdot \dot{s}^3\kappa\tau = \dot{s}^6\kappa^2\tau.$$

Step 3. Since $|\dot{\beta} \times \ddot{\beta}|^2 = \dot{s}^6\kappa^2$:

$$\tau = \frac{[\dot{\beta}, \ddot{\beta}, \ddot{\beta}]}{|\dot{\beta} \times \ddot{\beta}|^2}. \quad \square$$

Q.No. 5**Frenet–Serret Apparatus and Plane Curves**

(a) Compute the Frenet–Serret apparatus for $\beta(t) = (t - \cos t, \sin t, t)$.

Solution

Derivatives: $\dot{\beta} = (1 + \sin t, \cos t, 1)$, $\ddot{\beta} = (\cos t, -\sin t, 0)$, $\ddot{\beta} = (-\sin t, -\cos t, 0)$.

Speed: $|\dot{\beta}|^2 = (1 + \sin t)^2 + \cos^2 t + 1 = 3 + 2\sin t$.

Unit Tangent:

$$T = \frac{(1 + \sin t, \cos t, 1)}{\sqrt{3 + 2\sin t}}.$$

Cross product:

$$\dot{\beta} \times \ddot{\beta} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 + \sin t & \cos t & 1 \\ \cos t & -\sin t & 0 \end{vmatrix} = (\sin t, \cos t, -(1 + \sin t)).$$

$$|\dot{\beta} \times \ddot{\beta}|^2 = \sin^2 t + \cos^2 t + (1 + \sin t)^2 = 1 + (1 + \sin t)^2.$$

Binormal:

$$B = \frac{(\sin t, \cos t, -(1 + \sin t))}{\sqrt{1 + (1 + \sin t)^2}}.$$

Curvature:

$$\kappa = \frac{\sqrt{1 + (1 + \sin t)^2}}{(3 + 2\sin t)^{3/2}}.$$

Triple product for torsion:

$$\ddot{\beta} \times \ddot{\beta} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos t & -\sin t & 0 \\ -\sin t & -\cos t & 0 \end{vmatrix} = (0, 0, -1), \quad [\dot{\beta}, \ddot{\beta}, \ddot{\beta}] = \dot{\beta} \cdot (0, 0, -1) = -1.$$

Torsion:

$$\tau = \frac{-1}{1 + (1 + \sin t)^2}.$$

Principal Normal: $N = B \times T$.

(b) Prove β is a plane curve $\iff [\dot{\beta}, \ddot{\beta}, \ddot{\beta}] \equiv 0$ (given $\kappa \neq 0$).

Solution

Part 1 ($\Rightarrow$): Plane curve $\implies$ triple product = 0.

Step 1. Suppose β lies in a fixed plane with unit normal $\mathbf{n}$.

Then T and N always stay in that plane, so $B = \pm \mathbf{n} = \text{constant}$.

Step 2. Since B is constant: $B' = \mathbf{0}$. From the Frenet equation $B' = -\tau N$:

$$-\tau N = \mathbf{0} \implies \tau = 0.$$

Step 3. From part (iii) of Q4: $[\dot{\beta}, \ddot{\beta}, \ddot{\beta}] = \dot{s}^6 \kappa^2 \tau = \dot{s}^6 \kappa^2 \cdot 0 = 0$. ✓

Part 2 ($\Leftarrow$): Triple product = 0 $\implies$ plane curve.

Step 1. If $[\dot{\beta}, \ddot{\beta}, \ddot{\beta}] = 0$, then $\dot{s}^6 \kappa^2 \tau = 0$.

Since $\dot{s} \neq 0$ (regular curve) and $\kappa \neq 0$ (given), we must have $\tau = 0$.

Step 2. $\tau = 0 \implies B' = -\tau N = \mathbf{0}$, so $B = \mathbf{c}$ (a constant unit vector).

Step 3. Fix any point $\beta(t_0)$. Differentiate $(\beta(t) - \beta(t_0)) \cdot \mathbf{c}$:

$$\frac{d}{dt} [(\beta(t) - \beta(t_0)) \cdot \mathbf{c}] = \dot{\beta} \cdot \mathbf{c} = \dot{s} T \cdot B = 0 \quad (T \perp B \text{ always}).$$

Step 4. So $(\beta(t) - \beta(t_0)) \cdot \mathbf{c} = \text{const} = 0$. This means β lies in the plane:

$$\{p \in \mathbb{R}^3 : (p - \beta(t_0)) \cdot \mathbf{c} = 0\}. \quad \square$$